Drude Model

Discussion : 02/25/2016

$$\vec{D}(t) = \int_{-\infty}^{t} dt' \, \epsilon(t-t') \vec{E}(t') \; ; \quad \text{In freq. space, } \vec{D}(\omega) = \epsilon(\omega) \vec{E}(\omega)$$

Task : To find out $\epsilon(\omega)$ in metal.

Eq^n of motion : $m \dfrac{d^2x}{dt^2} + m\gamma_e v = -e\vec{E}$

External electric field, $\vec{E} = \vec{E_0} e^{-i\omega t}$, $q = -e$

$$\Rightarrow m\ddot{x} + m\gamma\dot{x} = qE_0 e^{-i\omega t}$$

Try $x = x_0 e^{-i\omega t}$ $\Rightarrow -x_0 m\omega^2 - m\gamma x_0 i\omega = qE_0$

$$\Rightarrow x_0 = \frac{-qE_0/m}{\omega^2 + i\omega\gamma}$$

$$\Rightarrow \boxed{x(t) = \frac{-qE_0/m}{\omega^2 + i\omega\gamma} e^{-i\omega t}}$$

Polarisation, $P = q\,x(t)N$, where N is the concentration

$$\vec{P} = \frac{-q^2\vec{E_0}/m \, e^{-i\omega t} N}{\omega^2 + i\omega\gamma} \quad , \quad \vec{D} = \epsilon_0\vec{E} + \vec{P}$$

$$\Rightarrow \vec{D} = \epsilon_0\vec{E_0}e^{-i\omega t} + \frac{-q^2\vec{E_0}/m \, e^{-i\omega t} N}{\omega^2 + i\omega\gamma}$$

$$\Rightarrow \vec{D} = \epsilon_0\vec{E_0}e^{-i\omega t}\left(1 - \frac{q^2N}{m\epsilon_0(\omega^2 + i\omega\gamma)}\right)$$

Thus, $\epsilon(\omega) = \epsilon_0\left(1 - \dfrac{q^2N}{m\epsilon_0(\omega^2 + i\omega\gamma)}\right) \Rightarrow \boxed{\epsilon(\omega) = \epsilon_0\left(1 - \dfrac{\omega_p^2}{(\omega^2 + i\omega\gamma)}\right)}$

$[\omega_p] \to \frac{1}{s}$, dimensions of frequency. where, $\omega_p^2 = \dfrac{q^2N}{m\epsilon_0}$, plasmon frequency

1) For $\omega < \omega_p$; ϵ real ; $\epsilon < 0 \to$ solutions decay exponentially in space.

2) For $\omega > \omega_p$; ϵ real ; $\epsilon > 0 \to$ solutions oscillatory, radiation can propagate.